

Real-Time End-to-End Purkinje Tree Generation and Fully Coupled Electrophysiology

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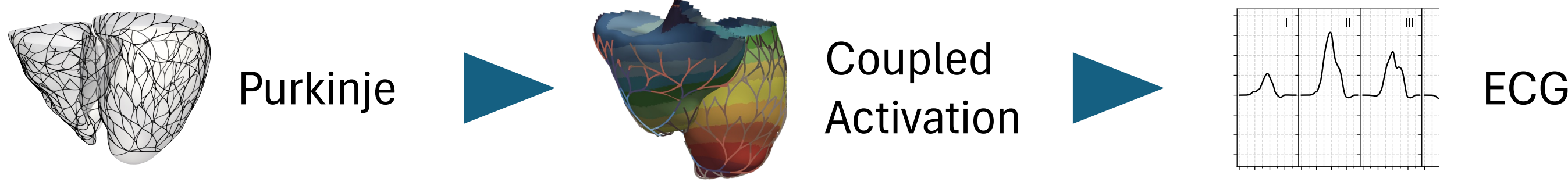
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Calibration of cardiac electrophysiology models requires **large numbers of model evaluations**. Even with eikonal models, end-to-end simulations still take **several seconds per run**.

Calibration may require 10^5 - 10^6 samples: this is **too slow**.

A key remaining bottleneck is the **Purkinje network**: generation, coupling and repeated solution.



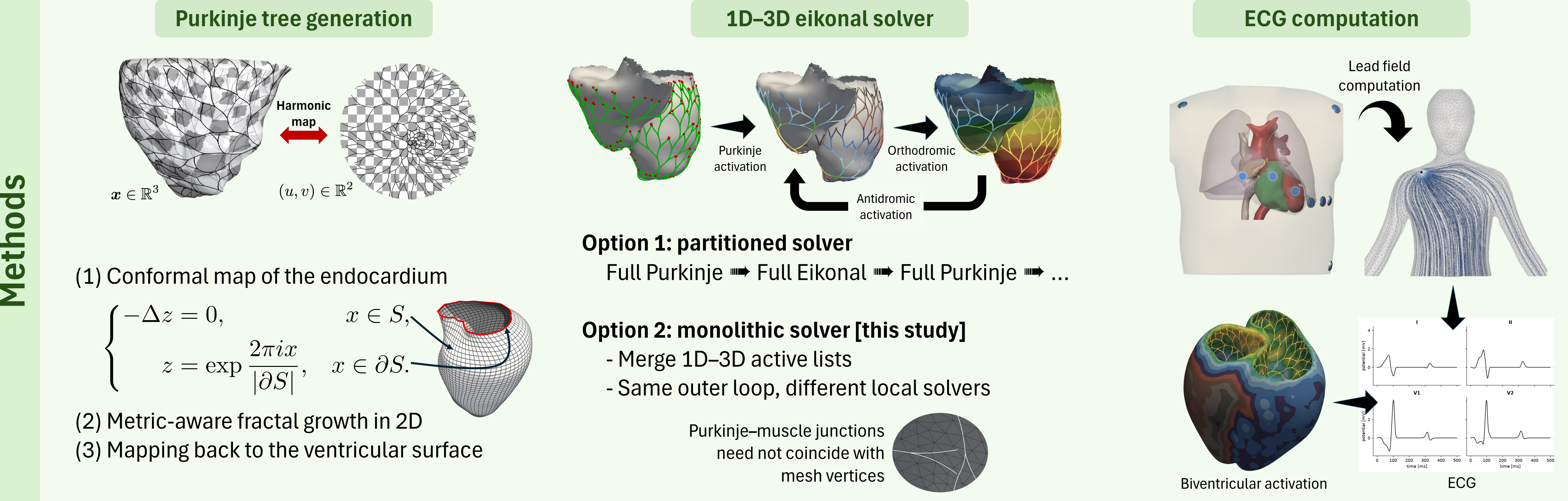
We target an **end-to-end real-time pipeline** for large-scale studies.

Methods

Implementation

Performance

Comparison to monodomain



| purky

```
from purky import Mesh, Parameters, generate_purkinje_cpp
endo = Mesh("endocardium.obj")
params = Parameters(init_node_id=198, second_node_id=186,
                    fascicles_angles = [-0.23, 0.23],
                    fascicles_length = [21.7, 5.795])
scale = (endo.uvscaling)**0.5
tree = generate_purkinje_cpp(endo.uv, endo.connectivity,
                             scale, params)
```

Key features:
C++ core • fast triangle queries • Python interface

| mish

```
from mish.models import FIMSolver, FIMCoupledSolver
fim_1L = FIMSolver(LVtree)
fim_1R = FIMSolver(RVtree)
fim_3d = FIMSolver(heart)
solver = FIMCoupledSolver(fim_3d, [fim_1L, fim_1R])
t_3d, [tL, tR] = solver.solve(x_pvc, t_pvc, [Lroot, Rroot], [[Lt, Rt]])
```

Key features:
GPU-ready via JIT • autodiff • multiple physics solvers

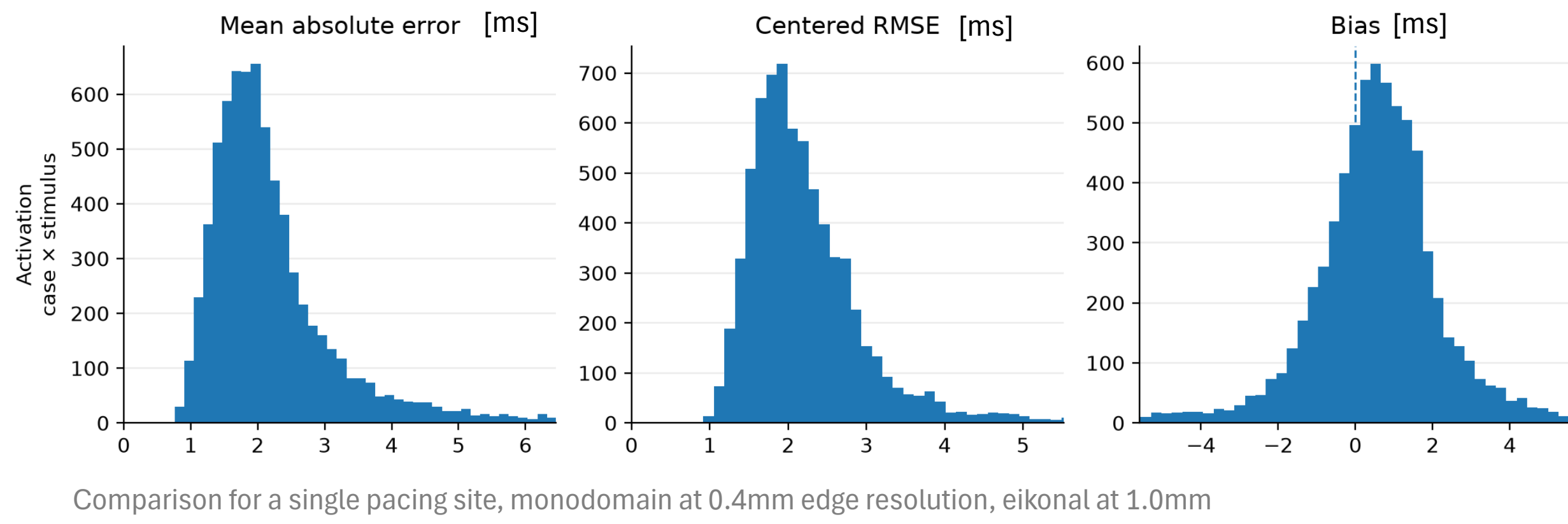
| forma

```
from forma import ShapeModel
from forma.ecg import ecg_from_activation
from forma.fit import fit_heart, fitted_mesh, points_from_mesh

anatomy = ShapeModel(sex="female").sample()
ecg = ecg_from_activation("act.vtu", anatomy=anatomy)

points = {"epi": epi_xyz, "lv_endo": lv_xyz, "rv_endo": rv_xyz}
fit = fit_heart(HeartModel(), points, n_modes=8)
```

Key features:
torso-heart atlas • fast sampling • fitting • automatic lead-field computation via CutFEM



Comparison for a single pacing site, monodomain at 0.4mm edge resolution, eikonal at 1.0mm

Application I

Probabilistic Calibration of a Closed-loop Cardiac EM

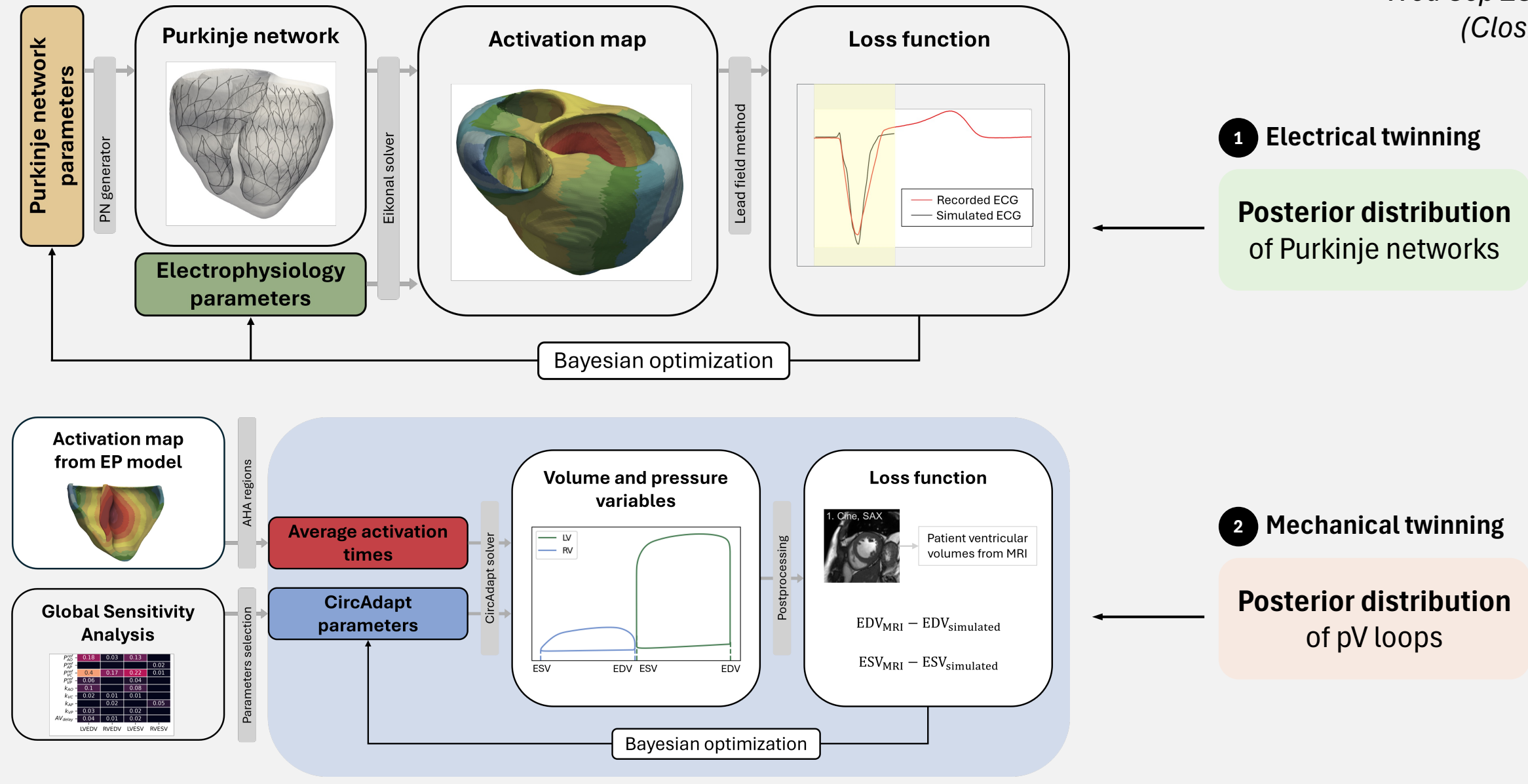
with E. Bergantin, F. Caforio, C. Augustin, F. Sahli Costabal

Bayesian personalization of the Purkinje network from ECG enables multiple activation patterns consistent with clinical data, providing electrophysiological **uncertainty quantification**.

Coupled electromechanics then propagates this uncertainty to hemodynamics, allowing **patient-specific comparison of CRT pacing strategies** at low computational cost.



Ester Bergantin
presentation!
Wed Sep 23 @ 3:30 PM
(Closing plenary)



Application II

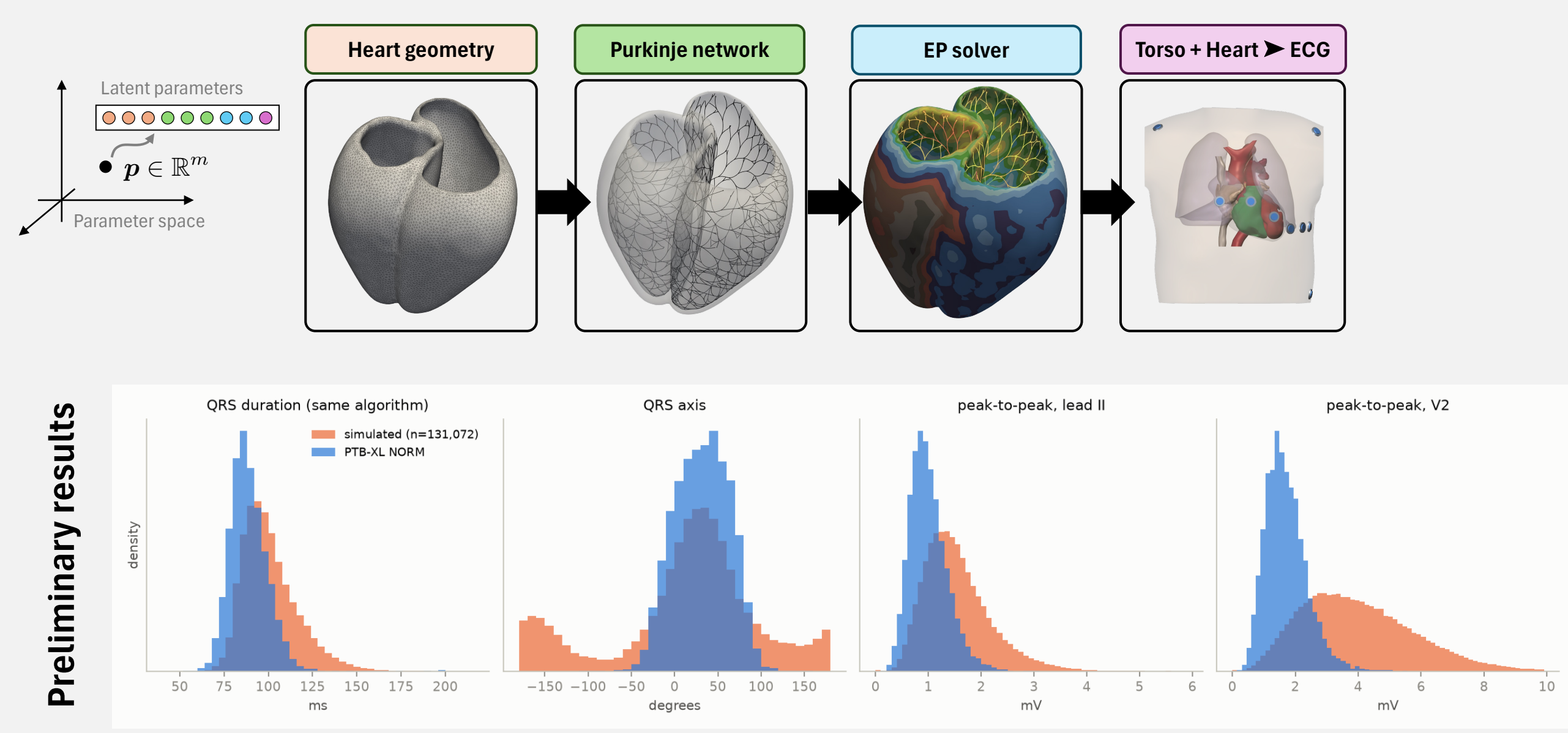
PurCA | Purkinje Component Analysis

with F. Sahli Costabal (and Alps Daint)

A **population-scale biventricular simulation atlas** combines PCA-based heart/torso geometry, Purkinje variability, activation and ECG generation in a hierarchical design.

The production workflow targets 4,096 anatomies × 32 EP/Purkinje realizations = **131,072 simulations**, reusing geometry-specific meshes and lead fields to keep the pipeline computationally tractable.

| purCA
 CSCS



References



• Schrotter T.G. et al. A physics-informed eikonal model for simulating arrhythmias in the human heart in real-time. *Nature Communications* **17**, 9209 (2026).



• Bergantin E. et al. Probabilistic calibration of a closed-loop cardiac electromechanical model with application to cardiac resynchronization therapy. *npj Digital Medicine* (2026).